

Lab 3 — Architectural Style Selection & Justification

Digital Campus Library Reservation Gateway | Sanjith KS | PES1UG24CS613 | Section K

Architecture Selection: We chose a **Layered Architecture** (Presentation → Business/Application → Data) for the Digital Campus Library Reservation Gateway System.

The external Payment Gateway and Email/SMS provider are reached only through adapter components in the business layer, never directly from the UI.

Two Reasons for the Choice

- 1. Queue correctness needs a single transactional boundary.** FR-001 (place hold in a FIFO waitlist), FR-002 (notify the next member within 5 minutes of check-in) and FR-005 (cancel/expire a hold and auto-promote the next member) all mutate the same waitlist. A layered design keeps this logic in one Reservation Manager over one ACID data store, so a queue position is assigned, promoted or expired atomically. Splitting these into independent services would force eventual consistency exactly where the requirement demands strict ordering.
- 2. The style matches the scale of the system and the team.** This is a single campus library with one deployment target, delivered by a small team across two sprints of 21 points each. Layering gives the separation of concerns the design needs — ten replaceable components, grouped into five subsystems and joined through published interfaces — without the operational cost of service discovery, distributed tracing and per-service pipelines that the scenario does not justify.

Security Advantage

Because every request travels downward through fixed layers, the data layer is unreachable from the presentation layer. Credentials enter through exactly one component, :AuthMemberService, which is therefore the only place the NFR-002 rule — lock the account for 15 minutes after 5 consecutive failed logins — has to be implemented and audited. Likewise, :PaymentService holds the only TLS 1.2+ link to the external gateway (GatewayTransaction); :MemberPortalUI and :ReservationManager never see card data, so the payment attack surface and audit scope shrink to a single component behind a single interface.

Performance Benefit

The layered split lets the read-heavy search path be separated from the transactional write path. :CatalogSearchEngine maintains its own inverted index, refreshed over the DataAccess interface, and answers paginated queries from that index instead of joining live loan tables — which is what makes NFR-001 (results over 50,000 titles in under 150 ms during peak enrolment) achievable while hold and renewal transactions continue against the data store. Calls between layers are in-process, so no per-request network hop is added to the busiest operation in the system.

Alternatives Considered and Rejected

- **Microservices:** attractive for scaling search and isolating payment failures, but rejected because the FIFO waitlist would be spread across service boundaries (data-consistency risk on the core requirement) and the operational overhead is disproportionate to one campus deployment.
- **Client-Server (thick client):** rejected because business rules — the 3-hold limit, the 2-renewal cap, the 48-hour pickup expiry — would live in the client, making them unenforceable and duplicated across the member and librarian clients.

Subsystems, Components and Interfaces in the Diagram

Presentation: :MemberPortalUI, :LibrarianConsoleUI. **Reservation Services:** :AuthMemberService, :ReservationManager, :CatalogSearchEngine. **Billing & Notification:** :PaymentService, :NotificationService. **Library Data Store:** :LibraryDataStore. **External Systems:** :PaymentGateway, :EmailSmsProvider.

Interfaces: MemberAccess, LibrarianAccess, AuthenticationServices, ReservationServices, AdminServices, MemberEligibility, CatalogSearch, PaymentProcessing, NotificationDispatch, ReceiptEvent, DataAccess, FineLedgerAccess, GatewayTransaction, MessageDelivery — each shown with a provided ball and a required socket.